

```
[4] ✓ 5s
# =====
# TELCO CUSTOMER CHURN PREDICTION
# =====

import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    classification_report
)

from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from xgboost import XGBClassifier

# =====
# LOAD DATA
# =====

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

# =====
# DATA CLEANING
# =====

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

# Remove customer ID
df.drop('customerID', axis=1, inplace=True)

# Target Encoding
```

Disk 65.52 GB available

Variables Terminal

27°C
Mostly sunny

Search

DELL

✓ 11:19

ENG
IN